

**RESEARCH
PAPER**

**Security Alert
Fatigue on
Cybersecurity
Operations**

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RESEARCH PAPER	1
Security Alert Fatigue on Cybersecurity Operations	1
1. Abstract:	2
2. Introduction:	2
2.1. Problem Statement:	2
2.2. Existing Problem:	2
2.3. Proposed Solution:	2
3. Literature Review:	2
3.1. References:	2
4. Methodology:	2
4.1. Use of Dataset	2
4.2. Data Preprocessing and Exploratory Data Analysis:	2
4.3. Machine Learning Model:	2
4.4. Model Evaluation:	2
4.5. Tools and Technologies:	2
5. Result and Discussion:	2
5.1. Performance Evaluation of Machine Learning Model	2
5.2. Impact on Security Alert Prioritization	2
6. Conclusion:	2
6.1. Future Work	2

1. Abstract:

The increasing volume of cybersecurity alerts generated by modern security systems has created a significant challenge for Security Operations Centres (SOCs), known as security alert fatigue. The excessive number of alerts, including false positives and low-priority events, can reduce the efficiency of security analysts and delay the detection of critical threats. This research proposes a machine learning-based intelligent security alert prioritization framework to address this challenge by automatically analysing and ranking security alerts according to their risk level.

The proposed approach utilizes Passive DNS and IP Geolocation datasets to extract meaningful network and behavioural features associated with suspicious activities. The methodology includes data preprocessing, exploratory data analysis, feature selection, machine learning model development, and performance evaluation. Machine learning algorithms are applied to classify security events and identify high-priority alerts while reducing unnecessary notifications.

The performance of the proposed framework is evaluated using standard classification metrics, including accuracy, precision, recall, F1-score, and confusion matrix analysis. The results demonstrate that machine learning techniques can effectively improve alert management by reducing alert overload and assisting security analysts in focusing on critical security incidents. This research highlights the potential of artificial intelligence-driven solutions in enhancing cybersecurity operations and improving threat detection efficiency.

2. Introduction:

Cybersecurity has become one of the most critical concerns for organizations due to the rapid growth of digital technologies, cloud computing, and interconnected networks. As cyber threats continue to evolve in complexity and frequency, organizations deploy multiple security solutions such as Intrusion Detection Systems (IDS), Intrusion Prevention Systems (IPS), Endpoint Detection and Response (EDR), and Security Information and Event Management (SIEM) platforms to monitor network activities. These security tools continuously generate alerts to identify suspicious behaviour and potential cyberattacks. However, the increasing volume of alerts has created a significant operational challenge known as **security alert fatigue**, where security analysts become overwhelmed by the number of alerts requiring investigation. This problem reduces operational efficiency, delays incident response, and increases the possibility of overlooking genuine threats. Artificial Intelligence (AI) and Machine Learning (ML) have emerged as promising approaches to automate alert analysis and prioritize high-risk events, enabling Security Operations Centres (SOCs) to respond more effectively to cyber incidents.

2.1. Problem Statement:

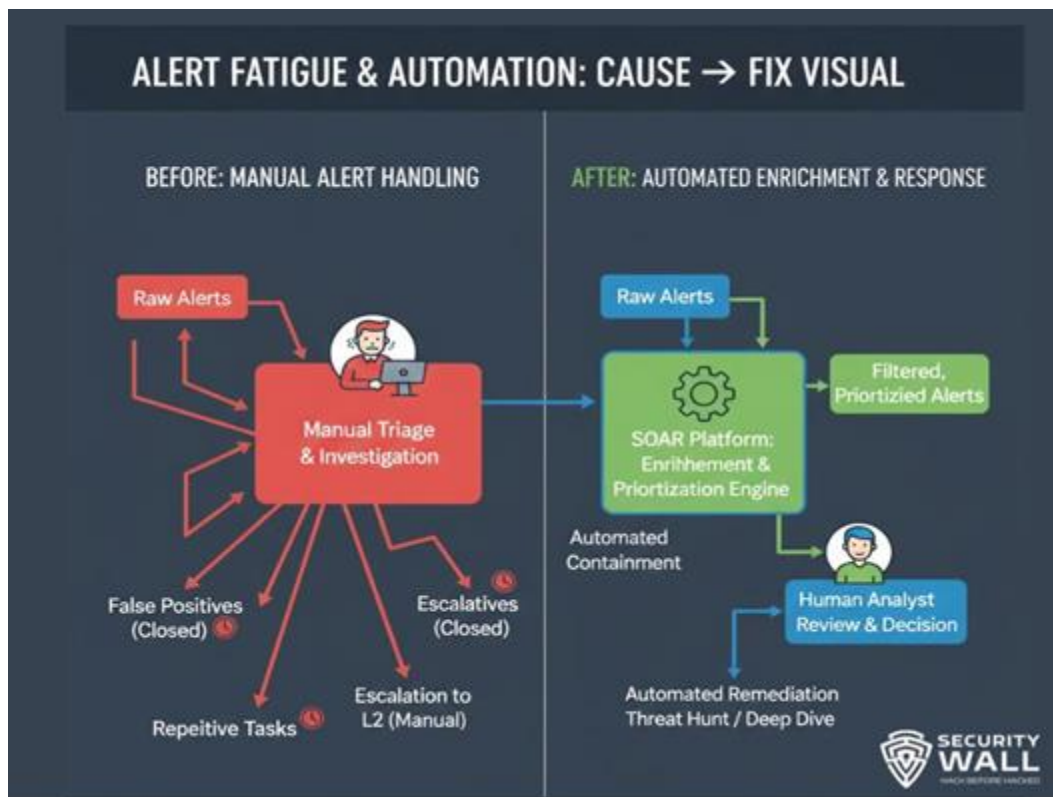
Modern Security Operations Centres (SOCs) receive thousands of security alerts daily from various monitoring tools. A substantial proportion of these alerts are false positives, duplicated notifications, or low-priority events that consume analysts' time and resources. Continuous exposure to excessive alerts results in cognitive overload, reduced analyst productivity, and delayed identification of critical security incidents. Consequently, organizations face increased operational costs and greater cybersecurity risks. Addressing security alert fatigue has therefore become an essential research challenge for improving SOC performance and strengthening organizational cyber defence.

2.2. Existing Problem:

Current security monitoring systems primarily rely on rule-based detection techniques and predefined thresholds to identify malicious activities. Although these methods successfully detect many threats, they often generate excessive alerts without considering contextual information or alert importance. Security analysts must manually investigate numerous alerts, many of which do not require immediate attention. This manual process is time-consuming, inconsistent, and difficult to scale in modern enterprise environments. Existing approaches also struggle to adapt to evolving attack techniques, leading to increased false-positive rates and reduced operational effectiveness. Consequently, important alerts may be delayed or overlooked during periods of high alert volume.

2.3. Proposed Solution:

This research proposes an AI-based intelligent security alert prioritization framework designed to reduce security alert fatigue within Security Operations Centres. The proposed framework integrates Passive DNS data with IP geolocation information to construct a comprehensive dataset describing network behaviour and contextual characteristics. Following data preprocessing, exploratory data analysis (EDA), and feature engineering, a Random Forest classifier is trained to categorize alerts into different priority levels. The framework aims to assist security analysts by automatically identifying higher-risk alerts, allowing them to focus on the most critical incidents while minimizing time spent investigating low-priority events.



3. Literature Review:

3.1. References:

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4. Methodology:

This research follows a machine learning-based methodology to develop an intelligent security alert prioritization framework for reducing security alert fatigue in Cybersecurity Operations Centres (SOCs). The proposed methodology focuses on analysing large volumes of security-related data and identifying important patterns that can help distinguish critical alerts from low-priority or false alerts. By applying data-driven techniques, the framework aims to support security analysts in managing alert overload and improving the efficiency of threat detection processes.

The methodology consists of several stages, including data collection, data preprocessing, feature extraction, machine learning model development, and performance evaluation. The collected cybersecurity datasets are prepared by handling missing values, removing inconsistencies, and transforming raw data into meaningful features. These features are then used to train machine learning models capable of classifying and prioritizing security alerts based on their risk level. Finally, the performance of the proposed approach is evaluated using standard evaluation metrics such as accuracy, precision, recall, F1-score, and confusion matrix to measure the effectiveness of the developed model.

4.1. Use of Dataset

The first stage of this research methodology involves the collection and utilization of cybersecurity datasets for developing an intelligent security alert prioritization framework. In this study, two datasets are used: Passive DNS data and IP Geolocation data. These datasets provide network-level information that helps in identifying patterns associated with suspicious activities and potential security threats.

The Passive DNS dataset contains historical domain and IP address relationship information, including domain behaviour, record frequency, similarity patterns, and other network-related characteristics. This information is useful for analysing malicious communication patterns and detecting abnormal activities. The IP Geolocation dataset provides additional contextual information such as geographical location, region, country, Autonomous System Number (ASN), and Internet Service Provider (ISP) details associated with IP addresses.

Both datasets are integrated to create a more comprehensive view of security events by combining behavioural and geographical features. The combined dataset helps the machine learning model understand different characteristics of network activities and assign appropriate risk priorities to security alerts. The use of these datasets enables the proposed framework to reduce unnecessary alerts, improve threat identification, and support security analysts in making faster and more accurate decisions.

4.2. Data Preprocessing and Exploratory Data Analysis:

After collecting and integrating the datasets, the next step of the methodology involves data preprocessing and exploratory data analysis (EDA). Data preprocessing is performed to improve the quality and reliability of the dataset before applying machine learning algorithms. The collected cybersecurity data may contain missing values, duplicate records, inconsistent formats, and irrelevant attributes that can affect model performance. Therefore, data cleaning techniques are applied, including

handling missing values, removing duplicate entries, converting data types, and selecting relevant features for further analysis.

The processed data is then analysed using exploratory data analysis techniques to understand the characteristics and distribution of security-related features. EDA helps identify important patterns, relationships, and anomalies within the dataset. Statistical analysis and visualization methods are used to examine feature distributions, identify unusual network behaviours, and understand the relationship between different attributes. This analysis provides valuable insights into the structure of the data and supports effective feature selection for the machine learning model.

Through preprocessing and EDA, the dataset is transformed into a structured and meaningful format, ensuring that the extracted features can effectively contribute to security alert classification and prioritization. This stage improves the accuracy and efficiency of the proposed framework by preparing high-quality data for model training and evaluation.

4.3. Machine Learning Model:

After data preprocessing and exploratory analysis, the next stage of the methodology focuses on developing machine learning models for security alert prioritization. The prepared dataset is divided into training and testing sets to enable effective model learning and performance evaluation. The training data is used to identify patterns and relationships between different security features, while the testing data is used to measure how accurately the model can classify unseen security events.

In this research, supervised machine learning techniques are applied to classify security alerts according to their risk level. The selected models analyse network-based features extracted from Passive DNS and IP Geolocation datasets to identify important indicators of malicious activities. Different classification algorithms can be evaluated to determine the model that provides better performance for distinguishing high-priority alerts from normal or low-risk events.

The developed machine learning model aims to reduce security alert fatigue by automatically prioritizing critical alerts and filtering less important events. This allows security analysts to focus their attention on potential threats instead of manually reviewing a large number of unnecessary alerts. The effectiveness of the model is evaluated using performance metrics such as accuracy, precision, recall, F1-score, and confusion matrix to determine its reliability in real-world cybersecurity environments.

4.4. Model Evaluation:

The final stage of the methodology involves evaluating the performance of the developed machine learning model. The evaluation process is performed to measure how effectively the model can classify and prioritize security alerts based on their risk level. The trained model is tested using unseen data to determine its ability to identify important security events and reduce unnecessary alerts.

Several performance evaluation metrics are used to analyse the effectiveness of the proposed approach, including accuracy, precision, recall, F1-score, and confusion matrix. Accuracy measures the overall correctness of the model, while precision and recall provide insights into the model's ability to correctly identify relevant security alerts. The F1-score combines precision and recall providing a balanced

performance measurement, especially important in cybersecurity scenarios where missing critical threats can have serious consequences.

The confusion matrix is also analysed to understand the classification results, including correctly identified alerts and misclassified events. Based on these evaluation results, the reliability and effectiveness of the proposed machine learning framework are determined. A successful model should achieve accurate alert prioritization while minimizing false positives, helping security analysts manage alert overload and improve threat response efficiency.

4.5. Tools and Technologies:

This research utilizes various tools and technologies for data processing, machine learning model development, and performance evaluation. Python is used as the primary programming language due to its extensive support for data analysis and machine learning tasks. Libraries such as Pandas and NumPy are used for data manipulation and numerical analysis, while Scikit-learn is used for implementing machine learning algorithms and evaluating model performance.

For exploratory data analysis and visualization, tools such as Matplotlib and Seaborn are used to identify patterns, distributions, and relationships within the cybersecurity datasets. The research experiments are performed using Google Collab, which provides a cloud-based environment for executing Python code and training machine learning models efficiently.

The datasets used in this study, including Passive DNS and IP Geolocation data, are processed and analysed using these technologies to develop an effective security alert prioritization framework. These tools support the complete research workflow, including data preprocessing, feature extraction, model training, testing, and evaluation.

5. Result and Discussion:

The developed machine learning framework was evaluated to determine its effectiveness in prioritizing cybersecurity alerts and reducing security alert fatigue. The experimental results demonstrate that the model was able to analyse network-based features extracted from Passive DNS and IP Geolocation datasets and identify patterns associated with different levels of security risk. The integration of both datasets improved the ability of the model to understand behavioural and geographical characteristics of network activities.

The results indicate that machine learning techniques can support Security Operations Centres (SOCs) by automatically distinguishing important alerts from less critical events. By reducing unnecessary alerts and highlighting high-risk activities, the proposed approach helps security analysts focus on threats that require immediate investigation.

5.1. Performance Evaluation of Machine Learning Model

The performance of the proposed model was evaluated using different classification metrics, including accuracy, precision, recall, F1-score, and confusion matrix analysis. Accuracy represents the overall correctness of the model, while precision measures how accurately the model identifies critical alerts. Recall evaluates the ability of the model to detect actual threats, which is an important factor in cybersecurity environments where missing a malicious event can cause significant damage.

The F1-score provides a balance between precision and recall, making it suitable for evaluating alert prioritization systems where both false positives and false negatives are important. The confusion matrix analysis further explains the classification behaviour of the model by showing correctly classified and incorrectly classified alerts. The evaluation results demonstrate that the proposed machine learning approach can effectively prioritize security alerts and reduce the workload caused by excessive false or low-priority notifications.

5.2. Impact on Security Alert Prioritization

The results of this research highlight the potential of machine learning-based approaches in improving cybersecurity alert management. By analysing multiple security-related features and assigning appropriate priority levels, the proposed framework helps reduce alert overload and improves the efficiency of security analysts. The ability to automatically identify high-risk events allows organizations to respond faster to potential threats while minimizing the time spent investigating irrelevant alerts.

Overall, the experimental findings show that the proposed framework provides an effective solution for addressing security alert fatigue by improving alert filtering, prioritization, and threat detection capabilities in modern cybersecurity environments.

6. Conclusion:

This research presented a machine learning-based security alert prioritization framework to address the problem of security alert fatigue in Cybersecurity Operations Centres (SOCs). The proposed approach utilizes Passive DNS and IP Geolocation datasets to analyse network characteristics and identify important patterns associated with potential security threats. Through data preprocessing, exploratory analysis, feature extraction, and machine learning techniques, the framework aims to classify and prioritize security alerts based on their risk level.

The results demonstrate that machine learning can effectively support security analysts by reducing the number of unnecessary alerts and improving the identification of critical security events. By automatically prioritizing high-risk alerts, the proposed framework helps minimize alert overload, improves response efficiency, and enables faster decision-making in cybersecurity environments. The evaluation process using classification metrics such as accuracy, precision, recall, F1-score, and confusion matrix confirm the potential of the proposed approach for improving security alert management.

Overall, this research highlights the importance of intelligent and automated alert prioritization systems in modern cybersecurity operations. The developed framework provides a practical solution for reducing security alert fatigue and enhancing the effectiveness of threat detection and response processes.

6.1. Future Work

Although the proposed framework provides an effective approach for security alert prioritization, several improvements can be explored in future research. Future work can focus on using larger and more diverse cybersecurity datasets collected from real-world Security Operations Centres to improve the generalization ability of the model.

Advanced machine learning and deep learning techniques, such as neural networks and ensemble learning approaches, can be investigated to improve classification performance and detect more complex attack patterns. Additionally, real-time alert processing and integration with Security Information and Event Management (SIEM) systems can be developed to enable automated prioritization in live cybersecurity environments.

Future research can also include explainable artificial intelligence (XAI) techniques to provide security analysts with understandable reasons behind each alert priority decision. This can improve trust in automated systems and support better human decision-making during threat investigation.

THE END